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India SMART UTILITY Week 2026

10 – 14 March 2026, New Delhi

Session : Name of the Session

P2P Energy Trading Platform

POWER CHANGE

Presented By

Dharmveer Singh, CEO-PowerXchange

About Powerxchange

PowerXchange is an energy-technology company building **India's first Energy Super App**, designed to offer multiple digital energy programs that empower consumers with greater choice and control over how they produce, buy, and use electricity.

One of its flagship initiatives is a large-scale Peer-to-Peer (P2P) energy trading platform—a digital,consumer-centric energy marketplace powered by AI and blockchain technologies. Built on the UnifiedEnergy Interface (UEI) architecture, the platform enables consumers and prosumers to directly buy andsell solar power in a transparent, secure, and near real-time settlement environment, while remainingfully aligned with existing DISCOM regulatory frameworks.

About Powerxchange

1. Powerxchange received the letter of award (LOA) from UPPCL for the implementationof P2P energy trading across discoms & Organisation deployed the solution.
2. Powerxchange was selected by REC Limited for Inter-state P2P Energy trading (As oneof the first use-case of India Energy Stack) pilot between Tata Power-DDL, BSES &PVVNL.
3. Powerxchange platform was demonstrated in front of Honourable Prime Minister ShriNarendra Modi in Global Impact AI Summit New Delhi on 16th Feb 2026.
4. Powerxchange platform was demonstrated in front of Honourable Power Minister ShriManohar Lal Khattar in Global Impact AI Summit New Delhi on 18th Feb 2026.

Why P2P energy trading?

PM Surya Ghar – World's Largest Residential Solar Program

RTS Installed till date:
03 M Approx

RTS target by 2027:
10 M approx

Application conversion rate : 22.7 %

Capacity target by 2027:
30 GW

Current Status
~3 million rooftop systems installed
~7 GW installed capacity added

Installation Velocity Gap to achieve 2027 targets

Metric	Value
Required pace	**583,000 installs/month**
Current pace	**~170,000 installs/month**

India is currently **tracking to miss the target by 60–70%**

THE VELOCITY PROBLEM — THE MATH IS BRUTAL

To reach **1 crore homes by March 2027**, India must complete roughly **583,000 installations every single month** from today. Current pace has averaged ~170,000/month over the life of the scheme. The scheme is delivering at **less than 30% of the required run-rate**. This is not a supply problem — solar panels are available and cheap. It is an **economic incentive problem**. Households don't install because the return on their ₹80,000–1,50,000 investment doesn't compute clearly enough.

RTS Solar Economics

The Net Metering Price Gap for Surplus Generation:

Households pay **₹6–12/kWh** to buy electricity
But earn only **₹2.25–3.65/kWh** when exporting solar power.
Note: Economically not viable for less income people living in rural and semi-urban area.

This **3–5× price gap destroys the ROI of rooftop solar.**

Model	Export Rate	Payback	Adoption
Net Metering	₹6–8	4–5 yrs	Strong but resisted by DISCOMs
Net Billing	₹2.25–3.65	7–10 yrs	Weak adoption
P2P Energy Trading	₹4.5–7	**3.5–5 yrs**	Strong adoption

Example Impact
Rural household (2kW solar):
•Net billing income: **₹270/month**
•P2P income: **₹600/month**

122% higher export revenue

Conclusion: Solar adoption is constrained by **low export monetization**.

The Solution: P2P Energy Marketplace

Peer-to-Peer Solar Trading Unlocks the Market

P2P platforms allow **households to sell excess solar directly to buyers** at market prices.

System-Level Benefits

Prosumers

•71–149% higher revenue

•Payback reduced to **3.5–5 years**

Consumers

•Buy solar power **20–35% cheaper**

Grid

•Due to local energy participants matching, grid stablises.

The Market Opportunity

Immediate Prosumer Base

3 million solar households

Stalled Pipeline

~7 million pending installations (Target of 10 Million solar homes by 2027)

- **Total Addressable Market**

100 million rooftop solar homes

India has built the **solar infrastructure and regulatory framework.**
What is missing is the marketplace layer that connects buyers and sellers at scale.

Regulatory Momentum for P2P Energy Trading in India

State-wise P2P Regulatory Guidelines:

- **Uttar Pradesh (UPERC, 2023):** Blockchain-based P2P energy trading regulations issued in 2023
- **Delhi (DERC, 2024):** P2P energy transaction guidelines released in 2024
- **Karnataka (KERC, 2024):** P2P solar transaction regulations introduced in 2024
- **Kerala (KSERC, 2025):** Guidelines supporting decentralized energy trading in 2025
- **Rajasthan (RERC):** P2P trading referenced under Virtual Net Metering (VNM) issued in 2025

National Developments

- **India Energy Stack (IES):** REC as nodal agency; interstate P2P trading pilot launched (PowerXchange partner)
- **Electricity Amendment Bill 2025 (Draft):** P2P energy trading recognized in proposed power market reforms

Key Insight:

Regulatory and policy frameworks across India are **actively enabling the adoption of P2P energy trading.**

PowerXchange Platform Key Features

AI-Driven Marketplace for Peer-to-Peer Energy Trading

PowerXchange is a digital marketplace that connects solar prosumers and energy buyers for secure, AI-driven peer-to-peer electricity trading.

Core Platform Capabilities

Secure Energy Settlement

- Blockchain-based energy transaction settlement
- Transparent and tamper-proof trade records

National Digital Infrastructure Ready

- Fully **India Energy Stack compliant**
- Supports regulated **interstate P2P energy transactions**

AI-Powered Trading Intelligence

- **Agentic AI trading workflow** with Voice-AI interface
- AI-based **demand forecasting and price discovery**

Smart Trade Negotiation Engine

- Automated **P2P trade negotiation** between buyers and prosumers
- Dynamic, market-driven energy pricing

Open Energy Marketplace

- **Beckn Protocol-enabled open network transactions**
- Interoperable ecosystem for energy buyers and sellers

Ecosystem Enablement

- Integration with **Urjamitra network** for field support and distributed energy participation

PowerXchange enables a transparent, intelligent, and interoperable digital marketplace for decentralized energy trading.

KEY TAKEAWAYS / RECOMMENDATIONS

Rooftop Solar Adoption Requires Better Economics
Low feed-in tariffs under current net billing models limit rooftop solar adoption and slow progress toward national solar targets.

Peer-to-Peer Energy Trading Unlocks Value
P2P trading enables market-based pricing, improving returns for solar prosumers while lowering energy costs for consumers.

Policy and Regulatory Momentum is Building
Multiple state regulators, national pilots, and emerging policy frameworks are enabling decentralized energy markets.

PowerXchange Enables the Energy Marketplace
PowerXchange provides a secure, AI-driven platform that connects buyers and sellers, facilitates price discovery, and settles transactions transparently.

Opportunity
With millions of rooftop solar systems expected in the coming years, **P2P energy trading platforms will become a critical infrastructure layer for India's decentralized energy economy.**

PowerXchange positions itself as the digital marketplace enabling scalable peer-to-peer energy trading in India.

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